

&" NuclearCraft

Plutonium Age

Complete Player Guide

Items & Materials · All Crafting Recipes · Machines
Radiation System · Zombies · Plutonium Titan Boss

PaperMC 1.21+ · Plugin Version 1.0 · Phases 1–10

All values configurable via YAML config files in /plugins/NuclearCraft/

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Material Substitution Reference

NuclearCraft custom items cannot be placed directly in vanilla crafting grids because Minecraft's recipe engine does not understand CustomModelData. Each custom item uses a specific vanilla material as a proxy in crafting. The plugin intercepts the crafting event and checks the real item via its CustomModelData tag before allowing the recipe to proceed.

&" Always obtain the custom item first (from gameplay or /nc give commands). A plain vanilla Echo Shard without the correct CMD tag will NOT work in recipes.

Crafting Grid Proxies

Vanilla Material (place in grid)	Custom Item It Represents	CMD
Echo Shard	Refined Plutonium Ingot	1103
Yellow Wool	Industrial Fabric	1315
Magma Cream	Radioactive Core	1101
Prismarine Crystals	Raw Plutonium Fragment	1102
Heart of the Sea	Irradiated Heart	1106
Pink Petals	Healing Petal	1105
Wheat Seeds	Mutated Seed	1104
Honey Bottle	Radiation Antidote (consumable)	1301
Glass Bottle	Radiation Serum (consumable)	1302

Block & Equipment Base Materials

Vanilla Base	What It Represents	CMD
Blast Furnace (block)	Nuclear Smelter (machine block)	1201
Smithing Table (block)	Nuclear Forge (machine block)	1401
Diamond Pickaxe	Radiation Drill (tool)	1108
Leather Helmet	Hazmat Helmet	1306
Leather Chestplate	Hazmat Chestplate	1307
Leather Leggings	Hazmat Leggings	1308
Leather Boots	Hazmat Boots	1309
Netherite Helmet	Plutonium Helmet	3001
Netherite Chestplate	Plutonium Chestplate	3002
Netherite Leggings	Plutonium Leggings	3003
Netherite Boots	Plutonium Boots	3004
Netherite Sword	Plutonium Sword	2002
Netherite Pickaxe	Plutonium Pickaxe	2001
Netherite Axe	Plutonium Axe	2003
Netherite Shovel	Plutonium Shovel	2004
Netherite Hoe	Plutonium Hoe	2005
Arrow (vanilla)	Plutonium Arrow (ingredient role)	—
Nether Star	Titan Core OR Reactor Heart (by context)	1107/1109
Amethyst Shard	Titan Fragment OR Mutated Crystal	1108/1111
Paper	Ancient Reactor Blueprint	1110

All Custom Items — Materials, Machines & Consumables

Raw Materials & Components

Item Name	Vanilla Base	CMD	Description
Raw Plutonium Fragment	Prismarine Crystals	1102	Mined from Plutonium Ore (Y -64 to -58) using Radiation Drill. 1–4 per ore with Fortune. Primary input for the Nuclear Smelter. Emits low radiation while in inventory.
Refined Plutonium Ingot	Echo Shard	1103	Output of the Nuclear Smelter (1 Fragment ! 1 Ingot, 15 seconds). Primary crafting material for all Plutonium weapons, tools, and armor. Also the repair material for Plutonium gear (+300 durability per Anvil use).
Radioactive Core	Magma Cream	1101	Dropped by Irradiated Zombies. Serves as fuel for the Nuclear Forge (2,000 energy per Core). Also ingredient in several recipes. High radiation source — store in Lead-Lined Crate.
Irradiated Heart	Heart of the Sea	1106	Rare drop from Irradiated Zombies. Required for MK-3 (1 Heart) and MK-4 (2 Hearts) Nuclear Forge upgrades. Also consumed in the Titan summoning ritual (1 Heart). Emits strong radiation.
Industrial Fabric	Yellow Wool	1315	Radiation-hardened woven material. Used to craft all four Hazmat Suit pieces. Also the repair material for Hazmat armor in an Anvil (+200 durability per repair).
Mutated Seed	Wheat Seeds	1104	Dropped by Irradiated Zombies. Plant ONLY on Radioactive Farmland (not normal farmland). Grows into Mutated Healing Plants through 5 stages. Stage 4 harvest yields Healing Petals.
Healing Petal	Pink Petals	1105	Harvested from fully grown Mutated Healing Plants (1–3 per harvest). Primary ingredient for Radiation Antidote and Radiation Serum. Right-clicking it directly gives Regeneration II for 5s and removes 25 radiation.

Machines & Tools

Nuclear Smelter

Blast Furnace · CMD: 1201

Industrial refining machine. Converts Raw Plutonium Fragments into Refined Plutonium Ingots (15 seconds each). Requires fuel. Minimum 500°C before processing begins. Emits +8 radiation per 5 seconds within 3 blocks while active. Right-click to open GUI.

Nuclear Forge

Smithing Table · CMD: 1401

Advanced equipment upgrade machine. Upgrades Plutonium tools/armor and Hazmat gear from base to MK-1, MK-2, MK-3, and MK-4. Each tier improves damage, speed, armor, and durability. Powered by Radioactive Cores (2,000 energy each). Right-click to open GUI.

Radiation Drill

Diamond Pickaxe · CMD: 1108

The ONLY tool that can safely mine Plutonium Ore. Mining ore with any other tool gives +25 radiation burst and drops nothing. 20% faster than Diamond Pickaxe (×1.2 speed modifier). 1,800 durability. Also has 10% chance to uncover extra Radioactive Debris while mining.

Consumables & Cures

Radiation Antidote

Honey Bottle · CMD: 1301

Right-click to consume instantly. Clears ALL radiation points, removes infection status, and removes all active radiation debuffs (Weakness, Slowness, Hunger, Nausea). Does NOT grant any immunity — you can be infected again immediately. Crafted with 2 Healing Petals + 1 Honey Bottle (shapeless).

Radiation Serum

Glass Bottle · CMD: 1302

Right-click to consume. Instantly clears ALL radiation and all debuffs (same as Antidote) AND grants 10 full minutes of radiation immunity — you cannot receive any radiation during this window. More expensive and rarer to craft than the Antidote.

All Custom Items — Equipment & Boss Items

Plutonium Tools & Weapons

Item	Base	CMD	Key Stats
Plutonium Sword	Netherite Sword	2002	+10 DMG · Speed 1.7/s · 2,500 dur · +10 rad/hit · Crit: +5 bonus rad · vs Stage 3+ +5 bonus rad
Plutonium Axe	Netherite Axe	2003	+11 DMG · Speed 1.0/s · 2,600 dur · 15% breaking time (2 block AcE +15 rad burst)
Plutonium Pickaxe	Netherite Pickaxe	2001	2,800 dur · 10% chance to uncover Radioactive Debris · Requires Radiation Drill for Plutonium Ore
Plutonium Shovel	Netherite Shovel	2004	2,500 dur · 15% chance to convert dug soil to Radioactive Farmland
Plutonium Hoe	Netherite Hoe	2005	2,000 dur · Fast crop tilling
Plutonium Arrow	Arrow (×4 output)	—	+25 rad on hit · Poison I 4s · Glowing 5s · Full charge crit: +20 bonus rad · Headshot
Radiation Drill	Diamond Pickaxe	1108	1,800 dur · ×1.2 mining speed · Only safe mining tool for Plutonium Ore

Plutonium Armor Set

All Plutonium Armor uses Netherite as base material. All four pieces share: Protection III + Unbreaking III enchantments. Each piece provides -15% radiation reduction.

FULL SET BONUS (all 4 pieces): Complete environmental radiation IMMUNITY + Speed II + Fire Resistance (permanent while worn). Set uses Netherite-tier stats.

Piece	Base	CMD	Armor	Toughness	Durability	Special
Plutonium Helmet	Netherite Helmet	3001	4	2	450	Night Vision
Plutonium Chestplate	Netherite Chestplate	3002	8	2	640	Resistance
Plutonium Leggings	Netherite Leggings	3003	6	2	590	Speed
Plutonium Boots	Netherite Boots	3004	4	2	480	Swim Speed

Hazmat Suit

Hazmat Suit uses Leather armor as base. Provides strong radiation protection without requiring Plutonium Ingots. Full set provides 80% radiation reduction.

FULL SET BONUS: 80% total radiation reduction (best non-plutonium radiation shield).

Piece	Base	CMD	Armor	Toughness	Durability	Special
Hazmat Helmet	Leather Helmet	1306	2	0.5	363	Radiation -20%
Hazmat Chestplate	Leather Chestplate	1307	5	0.5	529	Radiation -30%
Hazmat Leggings	Leather Leggings	1308	4	0.5	496	Radiation -20%
Hazmat Boots	Leather Boots	1309	2	0.5	430	Radiation -10%

Boss Items (Dropped by Plutonium Titan)

Titan Core	[Nether Star · CMD 1107]
4 Titan Cores consumed in summoning ritual. Extreme radiation. Can also be used in high-tier Phase 11 crafting.	
Titan Fragment	[Amethyst Shard · CMD 1108]
Drops from Titan on kill. Scales with contribution. Used in future high-tier crafting (Phase 11+). Extreme radiation.	
Reactor Heart	[Nether Star · CMD 1109]
10% chance per qualifying player. REQUIRED for MK-3 (1 Heart) and MK-4 (2 Hearts) Nuclear Forge upgrades.	
Mutated Crystal	[Amethyst Shard · CMD 1111]
8% chance per qualifying player. Used in future crafting (Phase 11+).	
Ancient Reactor Blueprint	[Paper · CMD 1110]
5% chance per qualifying player. Unlocks advanced Phase 11 reactor crafting. No current gameplay use — save it.	

Crafting Recipes — Machines & Blocks

All recipes crafted at a standard Crafting Table unless otherwise noted. Remember: proxy materials in grids must be the actual custom items with their CustomModelData tag.

Nuclear Smelter

D = Diamond O = Obsidian F = Furnace (vanilla)

Nuclear Smelter

Diamond	Obsidian	Diamond
Obsidian	Furnace	Obsidian
Diamond	Obsidian	Diamond

Result: 1× Nuclear Smelter block

Nuclear Forge

I = Echo Shard (= Refined Plutonium Ingot) D = Ancient Debris O = Obsidian S = Smithing Table (vanilla)

Nuclear Forge

Obsidian	Anc.Debris	Obsidian
Ref. Ingot	Sm. Table	Ref. Ingot
Obsidian	Anc.Debris	Obsidian

Result: 1× Nuclear Forge block | Ref. Ingot = Echo Shard proxy

Additional Craftable Blocks

Plutonium Block

P	P	P
P	P	P
P	P	P

P = Echo Shard (Refined Ingot)
Result: 1× Plutonium Block
(compact storage — emits more radiation than loose ingots)

Lead-Lined Storage Crate

Iron	Iron	Iron
Iron	Chest	Iron
Iron	Iron	Iron

Result: 1× Lead-Lined Crate
Stores fragments with ZERO radiation
leak to nearby players

Crafting Recipes — Plutonium Tools & Weapons

P = Echo Shard (= Refined Plutonium Ingot) S = Stick R = Magma Cream (= Radioactive Core)

Radiation Drill

Radiation Drill — Only safe mining tool for Plutonium Ore

R-Core	Diamond	R-Core
Diamond	Diamond	Diamond
	Stick	

R-Core = Radioactive Core (Magma Cream proxy) | 1,800 durability | ×1.2 mining speed

Plutonium Weapons

Plutonium Sword

	P	
	P	
	S	

+10 DMG · Speed 1.7/s
2,500 durability
+10 rad/hit · Crit: +5 bonus

Plutonium Axe

P	P	
P	S	
	S	

+11 DMG · Speed 1.0/s
2,600 durability
15% shockwave (3-block +15 rad)

Plutonium Tools

Plutonium Pickaxe

P	P	P
	S	
	S	

2,800 durability
10% debris find chance
Note: still needs Radiation Drill for Plutonium Ore

Plutonium Shovel

	P	
	S	
	S	

2,500 durability
15% soil !' Radioactive Farmland
on each dig

Plutonium Hoe

P	P	
	S	
	S	

2,000 durability
Fast crop tilling

Plutonium Arrow (×4 output)

	P	
	Arrow	
	Feather	

P = Echo Shard proxy
+25 rad on hit · Poison I (4s)
Glowing (5s) · Crit: +20 bonus rad

Crafting Recipes — Plutonium Armor

P = Echo Shard (= Refined Plutonium Ingot) in all slots shown.
All four pieces have: Protection III + Unbreaking III. Full 4/4 set grants environmental radiation IMMUNITY + Speed II + Fire Resistance.

Helmet & Chestplate

Plutonium Helmet

P	P	P
P		P

4 Armor · 2 Toughness · 450 dur
-15% radiation exposure
Passive Night Vision

Plutonium Chestplate

P		P
P	P	P
P	P	P

8 Armor · 2 Toughness · 640 dur
-15% radiation exposure
Passive Regeneration

Leggings & Boots

Plutonium Leggings

P	P	P
P		P
P		P

6 Armor · 2 Toughness · 590 dur
-15% radiation exposure
Passive Speed boost

Plutonium Boots

P		P
P		P

4 Armor · 2 Toughness · 480 dur
-15% radiation · Feather Fall II
NO ground-contact radiation

Repair

Repair material for Plutonium gear
Durability restored per ingot

Refined Plutonium Ingot (Echo Shard proxy) — use Anvil
+300 durability per ingot used in Anvil

Plutonium Armor Full Set Stats Summary

Piece	Armor	Toughness	Durability	Rad Reduction	Passives
Plutonium Helmet	4	2	450	-15%	Night Vision
Plutonium Chestplate	8	2	640	-15%	Regeneration
Plutonium Leggings	6	2	590	-15%	Speed
Plutonium Boots	4	2	480	-15%	Feather Fall II · No ground-rad
FULL SET BONUS	—	—	—	IMMUNE	Speed II + Fire Resistance (permanent)

Crafting Recipes — Hazmat Suit & Cure Items

W = Yellow Wool (= Industrial Fabric proxy). Hazmat Suit uses Leather armor as base.

Hazmat Helmet & Chestplate

Hazmat Helmet

W	W	W
W		W

2 Armor · 0.5 Tough · 363 dur
-20% radiation reduction
Leather base material

Hazmat Chestplate

W		W
W	W	W
W	W	W

5 Armor · 0.5 Tough · 529 dur
-30% radiation reduction
Best single Hazmat piece

Hazmat Leggings & Boots

Hazmat Leggings

W	W	W
W		W
W		W

4 Armor · 0.5 Tough · 496 dur
-20% radiation reduction

Hazmat Boots

W		W
W		W

2 Armor · 0.5 Tough · 430 dur
-10% radiation reduction

Hazmat Suit Repair & Full Set

Repair material for Hazmat Suit
Durability per repair
Full 4-piece set bonus

Industrial Fabric (Yellow Wool proxy) — use Anvil
+200 durability per Industrial Fabric used
-80% total radiation reduction

Radiation Antidote (Shapeless Recipe)

Shapeless = ingredients can be placed in ANY slots in the crafting grid, in any order.

Radiation Antidote — SHAPELESS (any arrangement)

Healing Petal	Healing Petal	Honey Bottle
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Result: 1× Radiation Antidote | Effect: Clears ALL radiation, infection & debuffs instantly | Does NOT grant immunity

Radiation Serum (Shaped Recipe)

H = Healing Petal R = Magma Cream (Radioactive Core) G = Gold Nugget A = Golden Apple B = Glass Bottle

Radiation Serum

Heal.Petal	R-Core	Heal.Petal
Gold Nugget	Golden Apple	Gold Nugget
Heal.Petal	Glass Bottle	Heal.Petal

Result: 1× Radiation Serum | Effect: Clears ALL radiation + grants 10 MINUTES of full radiation immunity
R-Core = Radioactive Core (Magma Cream proxy) | Golden Apple = vanilla Golden Apple

Nuclear Smelter Processing Recipe

Input	1× Raw Plutonium Fragment
Output	1× Refined Plutonium Ingot
Processing time	15 seconds (300 ticks at 20 TPS)
Required machine	Nuclear Smelter at "e500°C with fuel loaded

Radiation System — Stages & Sources

Radiation is measured in points from 0 to 1000. It accumulates through exposure to radioactive materials, zombie attacks, machines, and the Titan boss. As radiation rises, you progress through increasingly dangerous stages with worsening debuffs and eventual direct damage.

Radiation Stages

Stage	Name	Points	Debuffs & Effects
0	Healthy	0 – 99	No effects. Normal gameplay.
1	Minor Exposure	100 – 249	Weakness I · Nausea
2	Moderate Exposure	250 – 499	Weakness I · Slowness I · Nausea · 5% proximity spread
3	Severe Exposure	500 – 749	Weakness II · Slowness II · Hunger I · 0.5 HP damage per cycle · 15% spread
4	Critical Poisoning	750 – 1000	Weakness III · Slowness III · Hunger II · 1.0 HP damage per cycle · 25% spread · Death message on fatal damage

Natural Progression & Decay

Natural progression	While infected: Stage 1 +5pts/60s · Stage 2 +10pts/60s · Stage 3 +15pts/60s · Stage 4 +20pts/60s
Natural decay	Requires 10 minutes with NO new radiation !' -2 points per 60 seconds until clear

Complete Radiation Sources

Source	Radiation	Trigger / Interval
Plutonium Ore — within 1 block	+5 pts	Every 5 seconds passively
Plutonium Ore — within 2 blocks	+2 pts	Every 5 seconds passively
Plutonium Ore — within 3 blocks	+1 pt	Every 5 seconds passively
Mining without Radiation Drill	+25 burst	Per attempt — 5 second cooldown
Raw Plutonium Fragment in inventory	+1 pt	Every 30 seconds
Plutonium Block — within 1 block	+10 pts	Every 5 seconds
Plutonium Block — within 2 blocks	+5 pts	Every 5 seconds
Plutonium Block — within 3 blocks	+2 pts	Every 5 seconds
Radioactive Debris — within 5 blocks	+0.5/s	Continuous passive
Nuclear Smelter active — 3-block radius	+8 pts	Every 5 seconds
Radiation Cloud (on zombie death)	+5/s	10 seconds, 3-block radius — 20% spawn chance
Irradiated Zombie Lv.1 melee hit	+10 pts	Per hit
Irradiated Zombie Lv.2 melee hit	+20 pts	Per hit
Irradiated Zombie Lv.3 melee hit	+35 pts	Per hit
Alpha Zombie Lv.4 melee hit	+50 pts	Per hit
Irradiated Zombie aura (proximity)	+2 pts	Every 5 seconds within range
Titan Boss — Radiation Aura (passive)	+3/s	10-block radius, all phases continuously
Titan Boss — Titan Slam (Phase 1+)	+15 pts	Per slam hit on player — 5-block radius
Titan Boss — Radiation Wave (Phase 2+)	+40 pts	20-block radius burst — 15s cooldown
Titan Boss — Reactor Overload (Phase 3+)	+70 pts	18-block radius burst — 40s cooldown
Titan Boss — Energy Beam (Phase 3+)	+60 pts	Targeted player direct hit — 20s cooldown
Titan Boss — Nuclear Catastrophe (Ph.4)	+80 pts	30-block radius + sets all HP to 1 — 90s CD
Nuclear Forge Overload	+150 pts	8-block radius burst when Forge overloads
MK-IV Equipment Aura (enemy players)	+15/tick	Every 2 seconds within 3 blocks
Toxic Bloom (farming hazard)	+8 pts	Every 2 seconds, 4-block radius

Radiation System — Protection, Cures & Contagion

Armor Radiation Protection

Armor reduces all incoming radiation exposure by a flat percentage multiplier per piece worn.

Armor / Situation	Radiation Reduction	Notes
Any single armor piece (generic)	-15% per piece	Stacks: 4 pieces = up to -60% total
Hazmat Helmet	-20%	Leather base
Hazmat Chestplate	-30%	Best single piece for radiation protection
Hazmat Leggings	-20%	Leather base
Hazmat Boots	-10%	Leather base
Full Hazmat Suit (all 4 pieces)	-80% TOTAL	Best accessible radiation protection
Plutonium Helmet only	-15%	Netherite base
Plutonium Armor — full set (4/4)	FULL IMMUNITY	Zero environmental radiation while worn
Plutonium Boots special property	Ground radiation	Negates radioactive farmland & ground sources
Near Plutonium Ore with Hazmat Suit	×0.20 multiplier	80% of ore radiation blocked
Near Plutonium Ore with Plutonium Armor	×0.00 multiplier	Complete immunity to ore radiation zones

Cure & Immunity Summary

Cure Method	How to Use	Effect
Healing Petal (direct use)	Right-click to consume	Removes 25 radiation · Regeneration II for 5 seconds
Radiation Antidote	Right-click to consume	Clears ALL radiation, all debuffs & infection · No immunity granted
Radiation Serum	Right-click to consume	Clears ALL radiation & debuffs · 10 min full radiation immunity
Plutonium Armor (full set)	Wear all 4 pieces	Permanent immunity to all environmental radiation while worn
Natural decay (passive)	Avoid radiation for 10m	" 2 radiation per 60 seconds until Stage 0

Contagion — Radiation Spreading Between Players

Irradiated players at Stage 2 or higher can spread radiation to others through proximity, contact, or shared vehicles. Even healthy players standing near infected ones are at risk.

Spread Method	Condition	Radiation	Chance
Proximity — within 3 blocks	Infected at Stage 2	+25 pts	5% per 5 seconds
Proximity — within 3 blocks	Infected at Stage 3	+25 pts	15% per 5 seconds
Proximity — within 3 blocks	Infected at Stage 4	+25 pts	25% per 5 seconds
Physical contact — melee hit	Any stage	+50 pts	35% per hit
Shared vehicle (boat or minecart)	Any stage	+25 pts	20% per 15 seconds

Visual & Audio Indicators

Each radiation stage displays different coloured particles around the player:

Stage 1 — Minor	Bright green particles [RGB: 57, 255, 20]
Stage 2 — Moderate	Yellow particles [RGB: 255, 220, 0]
Stage 3 — Severe	Orange particles [RGB: 255, 100, 0]
Stage 4 — Critical	Dark red particles [RGB: 200, 0, 0]

A Geiger counter clicking sound effect plays and intensifies as your radiation level rises.

Nuclear Smelter — Complete Machine Guide

Overview

The Nuclear Smelter converts Raw Plutonium Fragments into Refined Plutonium Ingots — the primary crafting material for all Plutonium-tier equipment. It is an industrial machine that requires fuel and must reach a minimum operating temperature before processing begins.

Placement & Usage Steps

1. Craft the Nuclear Smelter (Diamond + Obsidian + Furnace — see Page 6)
2. Place it like any block in the world
3. Right-click the Smelter block to open the GUI
4. Insert Raw Plutonium Fragments in the INPUT slot
5. Insert fuel in the FUEL slot (Coal, Charcoal, Coal Block, Blaze Rod, or Lava Bucket)
6. The machine starts HEATING — wait for it to reach 500°C
7. Processing begins automatically once temperature is reached
8. Collect Refined Plutonium Ingots from the OUTPUT slot

Fuel Values

Fuel Item	Fuel Units	Processing Time (approx.)	Notes
Coal	100 units	~20 seconds	Basic fuel
Charcoal	80 units	~16 seconds	Slightly worse than Coal
Coal Block	900 units	~3 minutes	Very efficient per slot
Blaze Rod	120 units	~24 seconds	Slightly better than Coal
Lava Bucket	1,000 units	~3 min 20 seconds	BEST fuel per inventory slot

&" Lava Bucket gives the most processing time per inventory slot — ideal for long refining sessions.

Temperature System

Parameter	Value
Ambient temperature (idle, no fuel)	20°C
Minimum temperature to begin processing	500°C — machine heats up first before any processing
Maximum safe operating temperature	1,500°C
Heating rate (fueled, idle)	+5°C per tick cycle (every 0.2 seconds)
Active heating rate (while processing)	+2.5°C per tick cycle
Cooling rate (no fuel, idle)	" 1°C per tick cycle
Overheat duration	10,000 ms (10 seconds) before cooling begins
Overheat cooling rate	" 2°C per tick cycle (twice as fast as normal)
Restart threshold (after cooling)	80% of 500°C = 400°C (avoids full cold start delay)

Processing Recipe

Input	1× Raw Plutonium Fragment
Output	1× Refined Plutonium Ingot
Processing time	15 seconds (300 ticks at 20 TPS)
Fuel cost	1 fuel unit consumed per tick cycle while processing

Radiation Hazard

The Nuclear Smelter emits passive radiation to ALL players within 3 blocks while active.

Passive radiation amount	+8 radiation points per check
Check interval	Every 5 seconds (100 ticks)
Affected radius	3 blocks around the machine

&" Build the Smelter in an isolated room. Full Hazmat Suit reduces incoming radiation by 80% — at 8 points per check, you receive only ~1.6 pts/check with full Hazmat.

&" Maximum smelters per player: unlimited (configurable in smelter.yml).

Nuclear Forge — Complete Machine Guide

Overview

The Nuclear Forge upgrades Plutonium and Hazmat gear from base tier to MK-I, MK-II, MK-III, and finally MK-IV. Each upgrade tier adds percentage bonuses to damage, attack speed, armor, and durability. The Forge is powered by Radioactive Cores (each provides 2,000 energy).

Usage Steps

1. Craft and place a Nuclear Forge block (see Page 6 for recipe)
2. Right-click the Forge block to open the upgrade GUI
3. Place the item you want to upgrade in the ITEM slot
4. Insert Radioactive Cores in the FUEL slot (each = 2,000 energy)
5. Insert Refined Plutonium Ingots in the MATERIAL slot
6. For MK-3 and MK-4: also insert Irradiated Hearts in the HEART slot
7. Click the UPGRADE button — the process takes 3–12 seconds
8. On SUCCESS: item is upgraded to next tier
 - On FAILURE: item stays at current tier
 - On FAILURE at MK-4: 5% additional chance of item downgrading to MK-3

Upgrade Tiers — Full Requirements

Tier	Ingots	Cores	Hearts	Energy	Success%	Time	Stat Bonus	Fail Risk
MK-1	2	1	0	500	100%	3 sec	+5% all stats	None
MK-2	4	2	0	1,200	90%	5 sec	+10% all stats	None
MK-3	8	4	1	2,500	75%	8 sec	+20% all stats	None
MK-4	16	8	2	5,000	50%	12 sec	+35% all stats	5% downgrade ! MK-3

&" Ingots = Refined Plutonium Ingots (Echo Shard proxy) · Cores = Radioactive Cores (Magma Cream proxy)
&" Hearts = Irradiated Hearts (Heart of the Sea proxy) · All stats = damage, speed, armor, durability

MK-IV Special — Radiation Aura

Any player wearing at least one MK-IV upgraded piece gains a passive Radiation Aura.

Aura Parameter	Value
Aura radius	3 blocks around the player
Radiation applied to hostile mobs	+25 radiation per aura tick
Radiation applied to enemy players	+15 radiation per aura tick (PvP)
Aura tick rate	Every 2 seconds (40 ticks)
Affects friendly/team players?	No — only hostile mobs and enemies

Forge Energy System & Overload

Energy Parameter	Value
Energy per Radioactive Core	2,000 units
Maximum safe energy storage	10,000 units (5 cores max at once)
Energy decay rate (idle)	" 0.5 units per tick (prevents idle overload)
Overload threshold	>10,000 energy triggers overload
Overload shutdown duration	200 ticks (10 seconds) — Forge is disabled
Overload radiation burst	+150 radiation to all players within 8 blocks
Forge block destroyed on overload?	NO — block is safe, only radiation burst

&" Never insert more than 5 Radioactive Cores at once — that hits the 10,000 energy limit and triggers overload.

World — Ore & Mining

Plutonium Ore World Generation

Parameter	Value
Dimension	Overworld only (Normal environment)
Y-level spawn range	Y: " 64 to " 58 (deepest deepslate layer ONLY)
Spawn chance per chunk	60% of newly generated chunks contain a vein
Vein size	1–2 blocks per vein (deliberately rare)
Valid host blocks	Deepslate, Stone, Cobbled Deepslate, Tuff
Discovery alert	Players receive a chat notification when within 5 blocks
&" Plutonium Ore is significantly rarer than diamonds at this depth (~40% rarer). Explore deep cave systems at Y -60 or lower.	

Mining Rules

Scenario	Outcome
Mine with ANY tool EXCEPT Radiation Drill	+25 radiation burst immediately · Ore block does NOT drop anything
Mine with Radiation Drill (no enchants)	· 5-second cooldown before next burst 1× Raw Plutonium Fragment · 3–7 XP orbs · No radiation burst
Mine with Radiation Drill + Fortune I	1–2× Raw Plutonium Fragments
Mine with Radiation Drill + Fortune II	1–3× Raw Plutonium Fragments
Mine with Radiation Drill + Fortune III	1–4× Raw Plutonium Fragments
Mine with Radiation Drill + Silk Touch	1× Plutonium Ore block (no fragments)

Ore Radiation Zones

Standing near Plutonium Ore passively irradiates you every 5 seconds. Use proper protection.

Distance from Ore	Radiation per 5s	With Hazmat Suit	With Plutonium Armor
Within 1 block ("d1)	+5 pts	+1 pt (~80% blocked)	0 pts (IMMUNE)
Within 2 blocks ("d2)	+2 pts	+0.4 pts	0 pts (IMMUNE)
Within 3 blocks ("d3)	+1 pt	+0.2 pts	0 pts (IMMUNE)

Plutonium Block

Craft by placing 9× Refined Plutonium Ingots in a 3×3 crafting grid. Used for compact storage. Emits more radiation than individual ingots.

Plutonium Block Radiation	Points per 5s
Within 1 block of Plutonium Block	+10 pts per 5 seconds
Within 2 blocks	+5 pts per 5 seconds
Within 3 blocks	+2 pts per 5 seconds

&" Keep Plutonium Blocks in a sealed vault. Even full Hazmat Suit only reduces the nearest tier to +2 pts per 5s.

Radioactive Debris

Debris Parameter	Value
Location	Y: " 64 to " 32 underground (rare)
Rarity	0.1% spawn chance per chunk (10× rarer than Plutonium Ore)
Mining drop	15% chance to drop 1× Radioactive Core
Passive radiation	+0.5 radiation per second within 5 blocks (continuous)

Lead-Lined Storage Crate

Crafted from 8 Iron Ingots around a Chest in a standard Crafting Table (iron surrounds center chest). Stores Raw Plutonium Fragments and other radioactive items WITHOUT emitting any radiation to nearby players — unlike a normal chest which leaks +1 radiation per 10 seconds within 3 blocks.

Radioactive Farming & Cure System

Getting Started

Radioactive Farming is the primary way to produce Healing Petals — the ingredient for radiation cures. You need Mutated Seeds (from zombie drops) and Radioactive Farmland (created with a Plutonium Shovel).

Step-by-Step Farming Guide

STEP 1 — Get Mutated Seeds:

Kill Irradiated Zombies. All levels can drop Mutated Seeds (5% at Lv.1, up to 50% at Lv.4 Alpha).

STEP 2 — Create Radioactive Farmland:

Dig soil blocks (Dirt, Grass Block, Coarse Dirt) with a Plutonium Shovel.

Each dig has a 15% chance to convert the block to Radioactive Farmland.

Normal farmland rejects Mutated Seeds — you MUST use Radioactive Farmland.

STEP 3 — Plant Seeds:

Right-click to plant Mutated Seeds on Radioactive Farmland.

Attempting to plant on regular farmland shows an error message and cancels.

STEP 4 — Let It Grow:

Plants grow through 5 stages: 0 (seed just planted) !' 1 !' 2 !' 3 !' 4 (fully bloomed).

Radioactive Farmland gives +50% extra growth chance per random tick.

Vanilla Bone Meal works and forces growth stages normally.

STEP 5 — Harvest at Stage 4:

Right-click the fully grown plant (Stage 4 blossom).

Receive 1–3 Healing Petals and 0–2 bonus Mutated Seeds.

Receive 2–5 XP per harvest.

STEP 6 — Watch for Toxic Blooms:

1% chance a Stage 4 plant mutates into a Toxic Bloom instead of dropping normally.

Destroy it immediately with any tool — it drops nothing but stops irradiating.

Farming Stats

Parameter	Value
Growth stages total	5 stages (0 = planted, 4 = ready to harvest)
Radioactive Farmland growth bonus	+50% extra chance per random tick
Bone meal compatible	Yes — vanilla Bone Meal forces growth stages
Petals per harvest	1–3 Healing Petals
Seeds per harvest	0–2 Mutated Seeds (self-sustaining farm supply)
XP per harvest	2–5 XP orbs
Toxic Bloom mutation chance	1% — fully grown crop becomes a radiation hazard

Toxic Bloom — Farming Hazard

Toxic Bloom Parameter	Value
What it is	A mutated fully-grown crop that emits radiation instead of dropping petals
Radiation radius	4 blocks around the Toxic Bloom block
Radiation amount	+8 radiation points every 2 seconds
How to destroy	Break it with any tool (instantly) — drops nothing
Particle indicator	Green radiation particles visible around the block

&" Always harvest at Stage 4 as soon as possible — the longer you wait, the more time for a Toxic Bloom to appear.

Radioactive Farmland Radiation

Passive radiation

+1 radiation point every 2 seconds to nearby players

Plutonium Boots special

Completely negates ground-contact radiation from Radioactive Farmland

&" Wear Plutonium Boots while farming to cancel all ground radiation. Pair with Hazmat Suit top pieces for full protection.

Normal Chest vs Lead-Lined Crate (Storage Warning)

Container	Radiation Leak	Effect
Normal Chest (with fragments inside)	YES — leaks radiation	+1 radiation per 10s within 3 blocks
Lead-Lined Storage Crate	NO — zero leak	Completely safe storage for fragments

Irradiated Zombies — Levels, Loot & Events

Spawning Rules

60% of all naturally-spawning zombies become Irradiated Zombies. Baby zombies are excluded. Mob spawners do NOT produce Irradiated Zombies (configurable). Eligible spawn reasons: Natural, Jockey, Mount, Reinforcements, Village Defence, Chunk Generation.

Level Stats

Level (Spawn Weight)	HP	Damage	Speed Mult	KB Resist	Rad/Hit	XP Reward
Lv.1 — Standard (80%)	35 HP	5 DMG	×1.15	0.15	+10 pts	10 XP
Lv.2 — Enhanced (15%)	45 HP	7 DMG	×1.20	0.20	+20 pts	20 XP
Lv.3 — Powerful (4%)	60 HP	9 DMG	×1.25	0.30	+35 pts	40 XP
Lv.4 — Alpha (1%)	80 HP	12 DMG	×1.30	0.50	+50 pts	100 XP

&" Level 4 Alpha: displays a green Bossbar "&" Alpha Irradiated Zombie &" and a permanent Glowing effect. Visible from 20 blocks. Treat as a mini-boss encounter.

Loot Drop Chances

Drop Item	Level 1	Level 2	Level 3	Level 4 Alpha
Rotten Flesh	0–3 (always)	0–3 (always)	0–3 (always)	0–3 (always)
Radioactive Core (Magma Cream)	15%	25%	40%	100% (GUARANTEED)
Mutated Seed (Wheat Seeds)	5%	10%	20%	50%
Irradiated Heart (H.o.t.Sea)	1%	3%	7%	25%

Radiation Cloud (On Zombie Death)

Spawn chance per zombie death	20% chance
Cloud radius	3 blocks
Cloud duration	10 seconds
Radiation inside cloud	+5 radiation per second to players inside
Visual indicator	Green radiation particles fill the cloud area
&" Never stand on a freshly killed Irradiated Zombie — even a Level 1 cloud deals +50 radiation over 10 seconds.	

Night Events — Radiation Surge

Each in-game night has a 5% chance to trigger a Radiation Surge — a server-wide event that amplifies all radiation for the entire night.

Surge Parameter	Value
Trigger chance	5% per in-game night
Radiation effect	ALL radiation damage DOUBLED for the full night
Loot effect	ALL zombie drops DOUBLED for the full night
Player notification	Bossbar shown to all players: "&" RADIATION SURGE ACTIVE &"
Bossbar colour	Green
Start broadcast	"&" RADIATION SURGE! A wave of radioactive energy sweeps the world!"
End broadcast	"&" Radiation Surge has ended. The air clears..."

&" During a Radiation Surge, a Level 4 Alpha Zombie delivers +100 radiation per hit (doubled) and has a 50% Core drop rate and a 100% Irradiated Heart drop rate (doubled). Use Serums before venturing out on Surge nights.

Advancements

Advancement	Trigger	XP Reward
First Exposure	Receive radiation for the first time from any zombie	+50 XP
Mutant Hunter	Kill your first Irradiated Zombie	+100 XP
Core Collector	Collect your first Radioactive Core	+100 XP
Alpha Slayer	Kill your first Level 4 Alpha Zombie	+500 XP

Plutonium Titan — Complete Boss Guide

Summoning the Titan

REQUIREMENTS:

- 4× Titan Cores in your inventory (consumed on summoning)
- 1× Irradiated Heart in your main hand
- A completed Summoning Altar

ALTAR CONSTRUCTION:

Center block: Crying Obsidian
4 Corner blocks: Obsidian (placed exactly 2 blocks diagonally from the center, same Y level)
No other blocks are required — the 5-block altar is minimal.

SUMMONING STEPS:

1. Build the Summoning Altar as described above
2. Ensure 4× Titan Cores are anywhere in your inventory
3. Hold 1× Irradiated Heart in your main hand
4. Right-click the center Crying Obsidian block
5. All 4 Titan Cores are immediately consumed from your inventory
6. A 15-second summoning animation begins
7. The Plutonium Titan spawns at the altar center

COOLDOWN: 60 minutes between summons. Only one Titan may exist at a time.

Health Scaling

Titan health scales dynamically based on the number of players nearby at time of summon.

Players Nearby	Titan HP
1–3 players	5,000 HP
4–6 players	6,500 HP
7–10 players	8,500 HP
10+ players	10,000 HP
Base melee damage Melee speed	18.0 base damage · Follow range: 80 blocks · Melee range: 3.5 blocks Attacks every 1.2 seconds (24 ticks)

Combat Phases

Phase	HP Range	Speed	DMG	New Abilities Unlocked in This Phase
Phase 1	100% – 75%	×1.00	×1.00	Radiation Aura (passive) · Titan Slam
Phase 2	75% – 50%	×1.10	×1.20	+ Radiation Wave · Mutant Summoning
Phase 3	50% – 25%	×1.25	×1.50	+ Reactor Overload · Energy Beam
Phase 4	25% – 0%	×1.50	×2.00	+ Nuclear Catastrophe · Final Frenzy (one-time trigger)

All Titan Abilities

Ability	Active From	Details	Cooldown
Radiation Aura	Phase 1	+3 radiation per second to all players within 10 blocks — constant passive	None
Titan Slam	Phase 1	5-block radius · 12 base DMG · +15 radiation · knockback on targets	10 sec
Radiation Wave	Phase 2	20-block expanding ring · +40 radiation burst to all caught in ring	15 sec
Mutant Summoning	Phase 2	Summons 3 regular Irradiated Zombies + 1 Level 4 Alpha Zombie	25 sec
Reactor Overload	Phase 3	5-second charge animation !' 18-block radius explosion !' +70 radiation	40 sec
Energy Beam	Phase 3	Targets a random player · Deals 25 direct DMG + 60 radiation on contact	20 sec
Nuclear Catastrophe	Phase 4	10-second charge !' 30-block radius !' +80 radiation	90 sec
Final Frenzy	Phase 4	sets ALL nearby players to 1 HP Triggered once on entering Phase 4 — grants permanent +50% attack speed for rest of fight	Once

&" Nuclear Catastrophe: Retreat beyond 30 blocks during the 10-second charge animation. You survive with 1 HP but are set there regardless of health.

&" Mutant Summon spawns a Level 4 Alpha every 25 seconds from Phase 2 onward — prioritise killing them quickly or radiation accumulates rapidly.

Arena Hazard

Hazard radius	15 blocks from the Titan summon point
Arena radiation	+2 radiation per check within the hazard zone

Titan Kill Rewards

Reward	Condition	Amount / Chance
Server XP	Any contribution	5,000 XP total (distributed by contribution %)
Titan Fragments	Any contribution	1-8+ pieces (Scales with contribution %)
Refined Plutonium Ingots	Minimum 5% contribution	Scales with contribution percentage
Reactor Heart	Qualifying players	10% chance per player
Ancient Reactor Blueprint	Qualifying players	5% chance per player
Mutated Crystal	Qualifying players	8% chance per player
Loot Chest (at death loc.)	All participants	1× chest containing Cores, Fragments, Ingots
Kill Broadcast	Always on kill	"[player] has slain the Plutonium Titan!"

Combat System — Weapons, Mastery & Repair

Plutonium Weapon Radiation Effects

Weapon / Attack Type	Radiation Applied	Additional Effect
Plutonium Sword — base hit	+10 radiation	—
Sword — Critical hit (falling attack)	+15 radiation	+5 bonus radiation on crit
Sword — vs Stage 3+ victim	+15 radiation	+5 bonus when victim already infected
Plutonium Axe — base hit	0 base radiation	—
Axe — Shockwave (15% chance)	+15 AoE radiation	3-block radius AoE burst to all targets
Plutonium Arrow — direct hit	+25 radiation	Poison I (4 seconds) · Glowing (5 seconds)
Arrow — Critical (full bow charge)	+45 radiation	+20 bonus radiation on max draw
Arrow — Headshot bonus	+35 radiation	+10 bonus (approximated headshot)
MK-IV Aura — on mobs	+25/tick	Passive every 2 seconds within 3 blocks
MK-IV Aura — on enemy players (PvP)	+15/tick	Passive every 2 seconds within 3 blocks

Combo System

The combo system rewards sustained combat with bonus radiation on consecutive hits.

Combo Parameter	Value
Maximum combo stack	8 hits
Combo reset time	6 seconds after the last hit
Bonus radiation per stack	+5 extra radiation per hit ABOVE the first (hit 2 = +5, hit 3 = +10...)
Maximum combo radiation	+35 radiation bonus cap (regardless of stack count)

PvP Radiation Surge (Two Infected Players Fighting)

Surge Parameter	Value
Trigger condition	BOTH combatants must be at Stage 2 radiation or higher
Trigger chance	5% per hit during combat
Radiation to BOTH fighters	+ 30 radiation each
Radiation to bystanders	+ 15 radiation to players within 8 blocks of the duel
Cooldown between surges	15 seconds per pair

Weapon Mastery System

Each weapon type (Sword, Axe, Bow) gains mastery XP from hits and kills. Higher mastery tiers provide bonuses.

Mastery Level	XP Required	XP Gain per Action	Master Tier Bonus
Novice	0 XP	Starting level — no bonuses	—
Experienced	100 XP	Sword: +1/hit, +10/kill	—
Veteran	500 XP	Axe: +1/hit, +10/kill	—
Elite	1,500 XP	Bow: +2/hit, +15/kill	—
Master	4,000 XP	All weapons accumulate faster	+10% radiation on all weapon hits · +1.5 block axe shockwave radius

Equipment Repair

Equipment Type	Repair Material (Anvil)	Durability per Repair
Plutonium Sword/Axe/Pickaxe/Shovel/Hoe	Refined Plutonium Ingot (Echo Shard proxy)	+300 durability per ingot
Plutonium Helmet/Chestplate/Leggings/Boots	Refined Plutonium Ingot (Echo Shard proxy)	+300 durability per ingot
Hazmat Helmet/Chestplate/Leggings/Boots	Industrial Fabric (Yellow Wool proxy)	+200 durability per piece
Radiation Drill	Diamond (vanilla)	Standard diamond pickaxe repair rules

&" Use an Anvil to repair — place the item in the first slot and the repair material in the second slot.

Admin Commands & Configuration

All commands use /nuclearcraft or the short alias /nc. Commands require operator permissions (OP) unless otherwise configured in plugin.yml.

Radiation Commands

Command	Description
/nc radiation set <player> <0-1000>	Set a player's radiation to an exact value (0 clears all radiation)
/nc radiation add <player> <amount>	Add a specified number of radiation points to a player
/nc radiation clear <player>	Instantly clear ALL radiation from a player (equivalent to set 0)
/nc radiation status [player]	Display the target player's current radiation stage and exact point total

Item & Equipment Commands

Command	Description
/nc ore give fragment [amount]	Give Raw Plutonium Fragments to yourself
/nc ore give drill	Give a Radiation Drill
/nc equipment give sword	Give a Plutonium Sword
/nc equipment give axe	Give a Plutonium Axe
/nc equipment give pickaxe	Give a Plutonium Pickaxe
/nc equipment give shovel	Give a Plutonium Shovel
/nc equipment give hoe	Give a Plutonium Hoe
/nc equipment give helmet	Give a Plutonium Helmet
/nc equipment give chestplate	Give a Plutonium Chestplate
/nc equipment give leggings	Give a Plutonium Leggings
/nc equipment give boots	Give a Plutonium Boots
/nc equipment give hazmat-helmet	Give a Hazmat Helmet
/nc equipment give hazmat-chestplate	Give a Hazmat Chestplate
/nc equipment give hazmat-leggings	Give a Hazmat Leggings
/nc equipment give hazmat-boots	Give a Hazmat Boots
/nc farming give seed [amount]	Give Mutated Seeds
/nc farming give petal [amount]	Give Healing Petals
/nc farming give antidote [amount]	Give Radiation Antidotes
/nc farming give serum [amount]	Give Radiation Serums
/nc smelter give	Give a Nuclear Smelter block
/nc forge give	Give a Nuclear Forge block

Titan Boss Commands

Command	Description
/nc titan spawn	Force-spawn the Plutonium Titan at your current location
/nc titan kill	Instantly despawn the active Titan (gives no rewards)
/nc titan phase <1-4>	Force the Titan to a specific combat phase immediately

Server Commands

Command	Description
/nc reload	Reload ALL YAML config files without restarting the server
/nc help	Show all available commands and their syntax

Configuration Files

All values in this guide are configurable. Files are located in /plugins/NuclearCraft/ on the server.

Config File	Controls
config.yml	Global plugin enable/disable, debug mode
radiation.yml	Radiation stages, progression, decay, contagion, armor reduction, immunity
zombies.yml	Zombie spawn chance, level weights, stats, loot tables, radiation cloud, night events

ore.yml	Plutonium Ore generation, mining, radiation zones, debris, storage crate
smelter.yml	Smelter tick rate, temperature system, fuel values, processing time, radiation emission
forge.yml	Forge energy, upgrade tier requirements, success chances, MK-IV aura, overload
equipment.yml	Tool/weapon stats, armor stats, arrow stats, soil/debris mechanics, repair values
armors.yml	Plutonium armor stats, enchantments, set bonus, radiation reduction per piece
farming.yml	Seed growth stages, harvest drops, Toxic Bloom chance, cure durations
combat.yml	Weapon radiation values, combo system, PvP surge, weapon mastery levels/XP
titan.yml	Boss HP scaling, phase thresholds, ability stats, summoning cooldown, rewards
boss.yml	Additional boss configuration and balance values

&" Use /nc reload after editing any config file to apply changes without server restart.

NuclearCraft: Plutonium Age · Plugin Version 1.0 · PaperMC 1.21+ · Phases 1–10 Complete

